



Inferential Statistics - Solutions

Exercise 4.1. The data below is a random sample of the number of goals scored in 10 Premier league football matches taken from the first 3 weeks of the 2020/21 season.

0 3 1 0 6 4 1 1 3 2

- (a) Use the sample to estimate the mean number of goals likely to be scored throughout the season.

$\bar{x} = 2.1$, and this could be used as an estimate of μ .

- (b) Make at least 2 comments on the accuracy of the estimate.

As the population is unlikely to be Normal, the sample size is too small to give an accurate result. The number of goals in the first 3 weeks will not be typical of the whole season so the results may be unrepresentative.

Exercise 4.2.

- (a) For each of the methods described below, identify the type of sampling being used.

- (i) You assign a number to every employee in the company database from 1 to 1000, and use a random number generator to select 100 numbers.

Random Sampling

- (ii) All employees of the company are listed in alphabetical order. From the first 10 numbers, you randomly select a starting point: 6. From 6 onwards, every 10th person on the list is selected (6, 16, 26, 36, and so on), and you end up with a sample of 100 people.

Systematic sampling

- (iii) You collect data from everyone who sits on the desks near you in your office.

Convenience sampling

- (iv) You want to know more about the opinions and experiences of employees with a disability in the company, so you contact a sample of employees with support needs.

Judgemental sampling

- (v) You send out the survey to all staff at your company and many staff members decide to complete it.

Voluntary response sampling

- (vi) The company has 800 female employees and 200 male employees. You select 80 women and 20 men, to get a sample of 100 people.

Proportional stratified sampling

- (vii) You want to find out about the opinions of people who like playing board games at your company. You know someone in your office who plays board games regularly with other staff members, so you ask them to suggest others you could interview.

Snowball sampling

- (viii) The company has offices in 10 cities across the country (all with roughly the same number of employees in similar roles). You select 3 offices and choose a random sample from each.

Cluster sampling

- (b) A large number of spectators attend a football match at a ground in England. One of the stands contains 1390 seats, numbered from 1 to 1390.
- (i) Describe how random numbers could be used to select a random sample of 80 of these seats.
Select random numbers between 1 and 1390, ignoring repeats. Continue until 80 numbers are selected, then choose the corresponding seats.
 - (ii) The ground authority wishes to carry out a survey into spectators' opinions of the catering facilities at the ground. It proposes to ask the spectators occupying the 80 seats selected in part (i) to complete a questionnaire. Suggest two practical difficulties which might be encountered in carrying out this proposal.
e.g. The seat chosen may not have sold; e.g. the person sitting in the seat may refuse to participate
- (c) A market research company is employed to investigate the amount of interest taken in sport by the population of the United Kingdom. A new employee of the company, suggests that interviewers should be stationed outside each exit from the ground with instructions to interview every 100th person leaving the ground after the match.
- (i) State the name given to this type of sampling.
Systematic sampling
 - (ii) Give one reason why the results of the investigation suggested might be biased.
All the people leaving the ground have just attended the football match, so are probably football fans, so the sample may be biased.
 - (iii) Suggest two practical difficulties, other than the one you already mentioned, which might be encountered in carrying out these interviews.
e.g. If a lot of people leave the ground at the same time, it might be difficult to identify the 100th person. Also in a crowd, it is difficult to conduct an interview.

Exercise 4.3. (a) Let $X \sim N(34, 6^2)$. Let $\bar{X}$ be the sampling distribution of the mean of a random sample of 12 observations. Find the standard error of the mean.

Since $\sigma = 6$ and $n = 12$, the standard error is $\frac{6}{\sqrt{12}} = 1.73$

- (b) Let $X \sim N(23.4, 25)$. Let $\bar{X}$ be the sampling distribution of the mean of a random sample of 26 observations.
- (i) Find the standard error of the mean.
Since $\sigma = \sqrt{25} = 5$, the standard error is $\frac{\sigma}{\sqrt{n}} = \frac{5}{\sqrt{26}} = 0.981$
 - (ii) Write down the distribution of $\bar{X}$.
 $N(23.4, 0.981^2)$

Exercise 4.4. 1. Write down the z values required for

- (i) A 90% confidence interval
0.9495 (1.64) and 0.9505 (1.65) are equidistant from the value 0.95, so we can take 1.64 or 1.65. (Computation devices would give 1.64.)
 - (ii) A 98% confidence interval
2.33
 - (iii) A 99% confidence interval
0.9949 (2.57) and 0.9951 (2.58) are equidistant from the value 0.995, so we can take 2.57 or 2.58. (Computation devices would give 2.58.)
2. Let $X \sim N(\mu, 12^2)$. Find a 95% confidence interval for μ based on a sample of size 15 with mean 24.
(17.9, 30.1)
3. Let $X \sim N(\mu, 24^2)$. Find a 98% confidence interval for μ based on a sample of size 30 with mean 94.
(83.8, 104.2)

4. Let $X \sim N(\mu, 2.35^2)$. Find a 90% confidence interval for μ based on a sample of size 10 with mean 165.
(164, 166)